

Computer Networks



Overview of the Internet Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
 - World Wide Web and HTTP
 - FTP
 - Electronic Mail
 - DNS
- Peer-to-peer paradigm
 - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

① Introduction

- Providing Services
- Application-Layer Paradigms

② Standard Client-Server Applications

- World Wide Web and HTTP
- FTP
- Electronic Mail
- Domain Name System (DNS)

③ Peer-to-Peer Paradigm

- P2P Networks

Peer-to-Peer (P2P) Paradigm

Definition

A **Peer-to-Peer (P2P)** network is a distributed network where each computer (peer) acts as both a client and a server, sharing its own resources directly with other peers.

Characteristics

- No dedicated server is required.
- Every peer can provide and receive services.
- Resources such as files, storage and bandwidth are shared.
- Communication occurs directly between peers.

Key Idea

Unlike the Client–Server model, every peer has equal responsibility.

Evolution of Peer-to-Peer Networks

Historical Development

- **1987** – WWIVnet (Wayne Bell)
- **1999** – Freenet (Ian Clarke)
- **1999–2001** – Napster (Shawn Fanning)
- **2000** – Gnutella
- **2001** – FastTrack (Kazaa)
- **2001** – BitTorrent
- **2001** – WinMX
- **2001** – GUNet

Exam Point

Napster popularized Peer-to-Peer file sharing, although it was later closed due to copyright violations.

Working of a P2P Network

How P2P Works

- 1 A peer joins the network.
- 2 The peer shares one or more files.
- 3 Other peers search for required files.
- 4 Interested peers download the file directly.
- 5 Downloaded files are again shared with others.
- 6 More downloads create more copies, increasing availability.

Challenge

Since peers frequently join and leave the network, the system must continuously keep track of

- Active peers
- Shared files
- File locations

Types of Peer-to-Peer Networks

Based on the organization of peers, P2P networks are classified into:

① Centralized P2P Networks

- Uses a central directory server.
- File transfer occurs directly between peers.

② Decentralized P2P Networks

- No central directory.
- Peers organize themselves.
- Can be
 - Unstructured
 - Structured

Exam Point

P2P Networks are broadly classified into **Centralized** and **Decentralized** networks.

Centralized P2P Networks

Definition

A Centralized P2P Network uses a central server only for maintaining the directory of peers.

Working

- Peer registers with the central server.
- Peer provides
 - IP Address
 - List of shared files
- Central server maintains the directory.
- Actual file transfer occurs directly between peers.

Key Idea

Directory Management follows the Client–Server model, while file transfer follows the Peer-to-Peer model.

Napster – A Hybrid P2P Network

Napster

Napster is the best-known example of a **Centralized (Hybrid) Peer-to-Peer Network**.

Operation

- 1 Peer registers with Napster server.
- 2 Server stores
 - Peer IP address
 - List of shared files
- 3 Another peer searches for a file.
- 4 Server returns peers having the file.
- 5 File is downloaded directly from one peer.

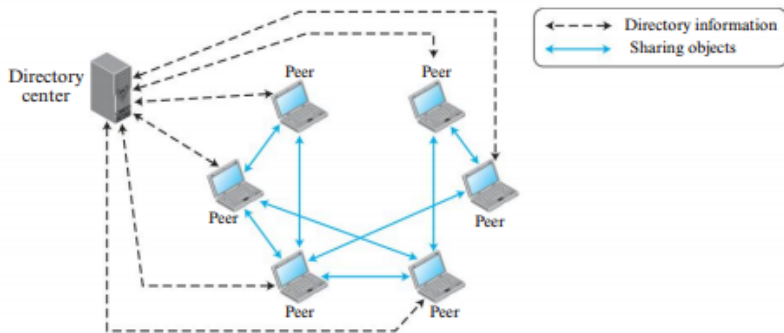
Important

Napster used multiple directory servers to reduce overload.

Directory Server Responsibilities

- Maintains peer list
- Stores shared file information
- Responds to search queries
- Continuously updates peer information

Figure 2.46 *Centralized network*



Advantages and Disadvantages of Centralized P2P

Advantages

- Simple directory management
- Fast file searching
- Easy peer discovery
- Easy maintenance

Disadvantages

- Heavy traffic at server
- Single point of failure
- Vulnerable to attacks
- Legal issues (Napster copyright case)

Exam Point

The central directory was the major reason for Napster's shutdown.

Decentralized P2P Networks

Definition

A **Decentralized P2P Network** does not use a central directory server. Peers organize themselves into an **overlay network**.

Overlay Network

- Logical network built on top of the physical Internet.
- Peers communicate directly.
- No single point of failure.

Types

- Unstructured Networks
- Structured Networks

Exam Point

Decentralized P2P eliminates the dependency on a central server.

Unstructured P2P Networks

Definition

Nodes are connected randomly without following predefined rules.

Characteristics

- Random node connections
- No central directory
- Search requests are flooded through the network
- Generates high network traffic
- Search may fail even if the file exists

Examples

- Gnutella
- Freenet

Gnutella

A decentralized and unstructured Peer-to-Peer network.

Working

- 1 Node A requests a file.
- 2 Request is sent to a neighbouring node.
- 3 If neighbour knows the file location, it returns the peer address.
- 4 Otherwise, it forwards (floods) the request to all its neighbours.
- 5 Once the file owner is found, Node A downloads the file directly.

Key Point

File transfer occurs using protocols such as HTTP after locating the peer.

Bootstrap, Ping and Pong

Bootstrap

- When Gnutella software is installed, it contains a list of known peers.
- These peers become the initial neighbours.

Ping Message

- Sent to check whether a neighbour is alive.

Pong Message

- Reply sent to indicate that the neighbour is active.

Exam Point

Bootstrap helps a new peer join the network by discovering its first neighbours.

Flooding and Improving Scalability

Problem: Flooding

- Search requests are forwarded to many peers.
- Produces excessive traffic.
- Reduces scalability.

Solutions in Modern Gnutella

- Ultra Nodes
 - Perform routing.
- Leaf Nodes
 - Share files but do not route queries.
- Query Routing Protocol (QRP)
- Dynamic Querying (DQ)

Key Idea

Ultra Nodes, QRP and Dynamic Querying reduce network traffic and improve scalability.

Structured P2P Networks

Definition

A **Structured P2P Network** follows predefined rules to organize peers, allowing files to be located efficiently.

Characteristics

- Organized network structure
- Efficient searching
- Better scalability
- Lower network traffic
- Faster file lookup

Technique Used

- Distributed Hash Table (DHT)

Distributed Hash Table (DHT)

Distributed Hash Table (DHT)

DHT is a distributed lookup technique that maps a key to the peer storing the corresponding data.

Advantages

- Fast searching
- Efficient routing
- Scalable
- Reduced flooding
- Distributed storage



Applications

- Distributed Data Structures (DDS)
- Content Distribution Systems (CDS)
- Domain Name System (DNS)
- Peer-to-Peer File Sharing

BitTorrent

BitTorrent is one of the most popular Structured Peer-to-Peer file sharing protocols.

Features

- Uses Distributed Hash Table (DHT)
- Efficient file discovery
- High scalability
- Reduced network overhead
- Faster downloads

Key Point

Unlike Gnutella, BitTorrent avoids flooding by using DHT.

Comparison of P2P Networks

Feature	Centralized	Unstructured	Structured
Directory	Central Server	Distributed	Distributed
Search	Fast	Flooding	Efficient Routing
Scalability	Moderate	Poor	High
Traffic	Moderate	High	Low
Examples	Napster	Gnutella, Freenet	BitTorrent

Exam Point

Structured P2P networks provide the best scalability and search efficiency.

Summary and Key Takeaways

Key Takeaways

- P2P Paradigm
- Evolution of P2P
- P2P Working
- Centralized P2P
- Napster
- Decentralized P2P
- Gnutella
- Flooding
- Ultra Nodes
- DHT
- BitTorrent
- P2P Comparison

Important Exam Points

- Difference between Client–Server and Peer-to-Peer
- Centralized vs Decentralized P2P
- Napster Architecture
- Gnutella Search Process
- Flooding and Scalability